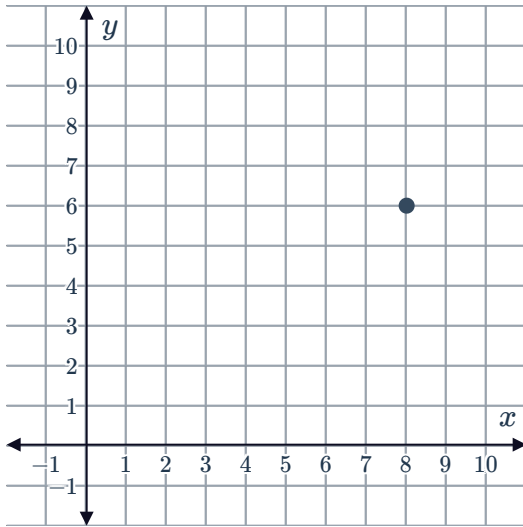


Translations



1

a

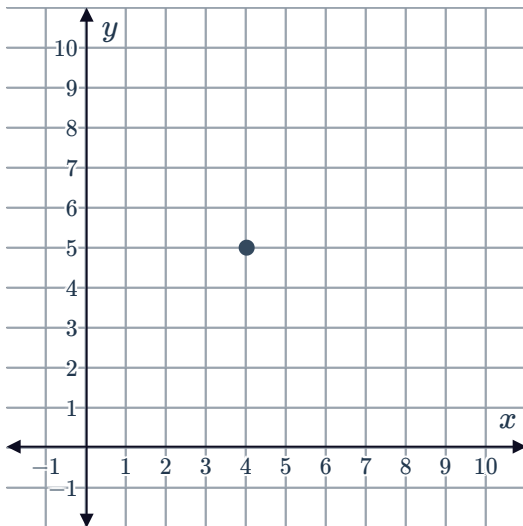


b (8,4)

c (2,4)

2

a



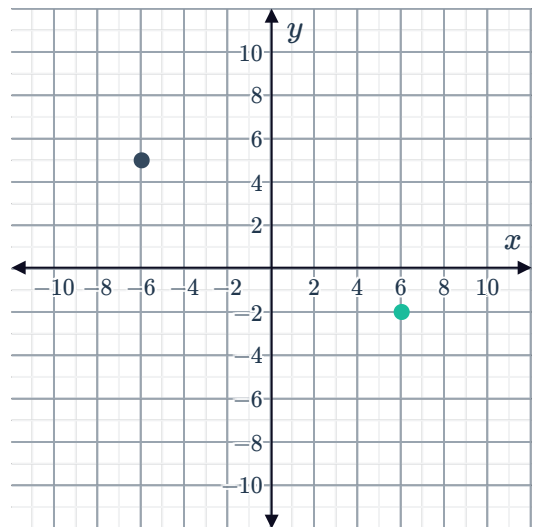
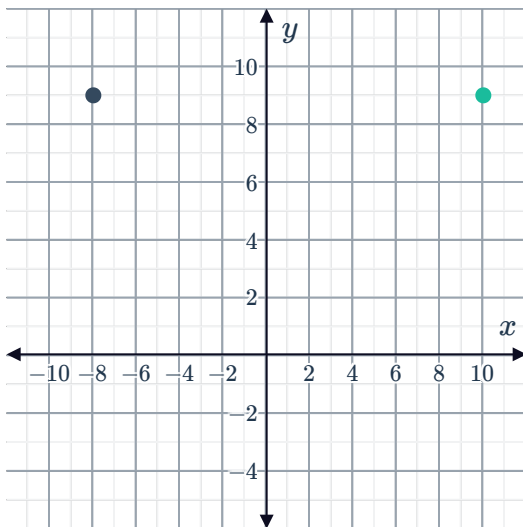
b (4,7)

c (6,7)

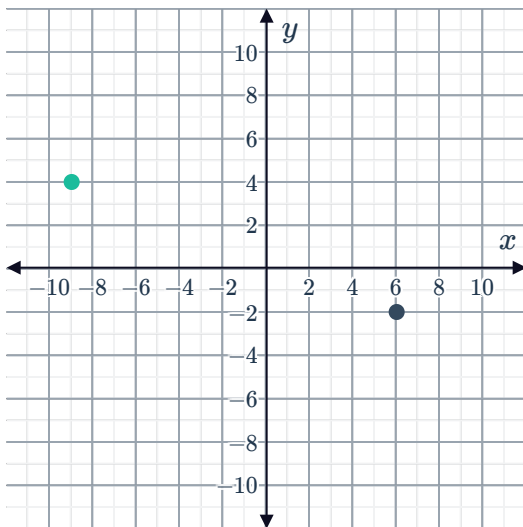
3



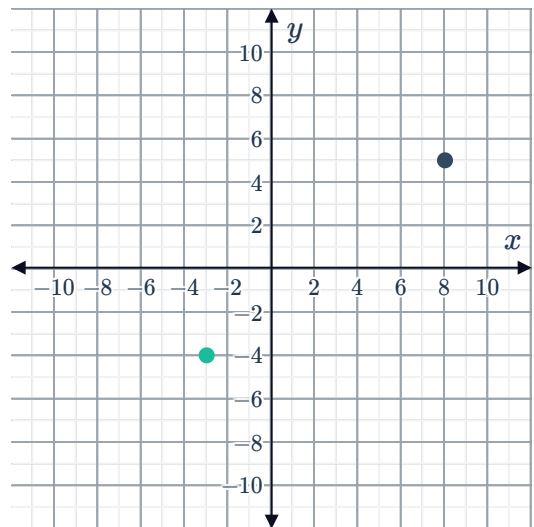
a



c



d



4

- a Trapezoid $ABCD$ is translated 7 units left to produce trapezoid $A'B'C'D'$.
- b Triangle ABC is translated 10 units right to produce triangle $A'B'C'$.
- c Polygon $ABCDEFG$ is translated 8 units right to produce polygon $A'B'C'D'E'F'G'$.
- d Trapezoid $ABCD$ is translated 8 units down to produce trapezoid $A'B'C'D'$.
- e Triangle ABC is translated 7 units up to produce triangle $A'B'C'$.
- f Polygon $ABCDEFG$ is translated 11 units up to produce polygon $A'B'C'D'E'F'G'$.
- g Trapezoid $ABCD$ is translated 9 units left and 3 units up to produce trapezoid $A'B'C'D'$.
- h Triangle ABC is translated 11 units left and 8 units up to produce triangle $A'B'C'$.

5 $(-3, 5)$

6 $(-7, 4)$



7

a $(8, -4)$

b $(-2, -4)$

c $(3, 1)$

d $(3, -9)$

8

a 3 units right

b 4 units left

c 4 units down

d 2 units up

e 7 units up

f 5 units right and 3 units up

g 4 units right and 1 unit down

9

a 2 units right and 1 unit up

b 6 units left

10 *B* or *D*

Additional questions

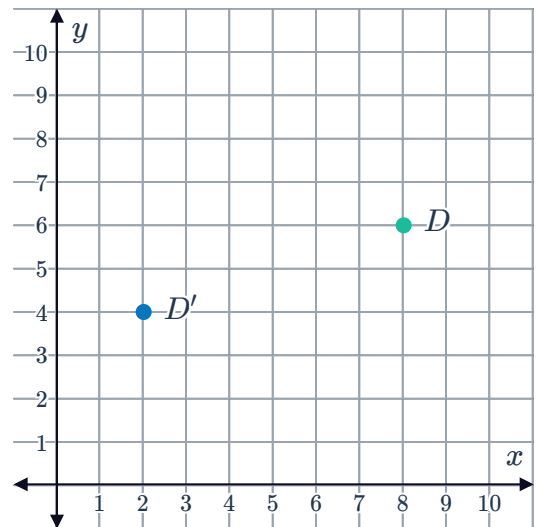
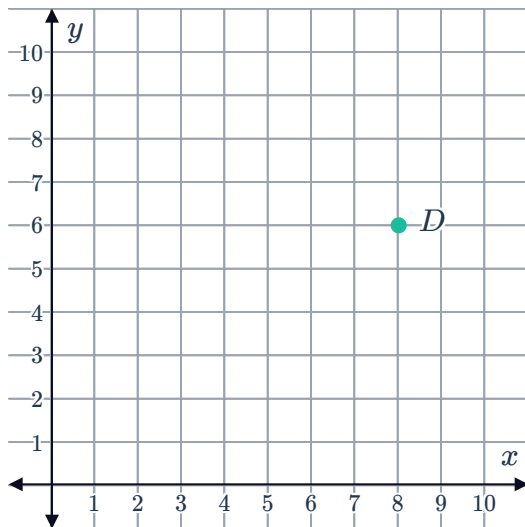
11 The figure remains the same shape and size, but is shifted 4 units to the left.

12

- a The graph does not show a translation, since $ABCD$ and $A'B'C'D'$ are different sizes and transformations do not change the size of a shape.
- b The graph shows a translation, since $ABCD$ and $A'B'C'D'$ are the same size and the image is shifted down and to the left of the preimage.

13

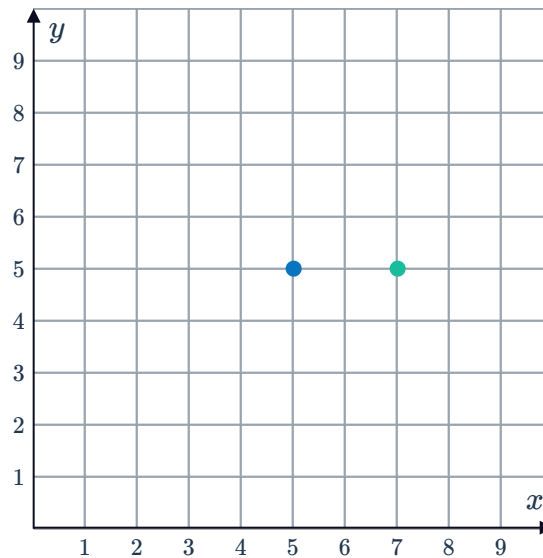
a



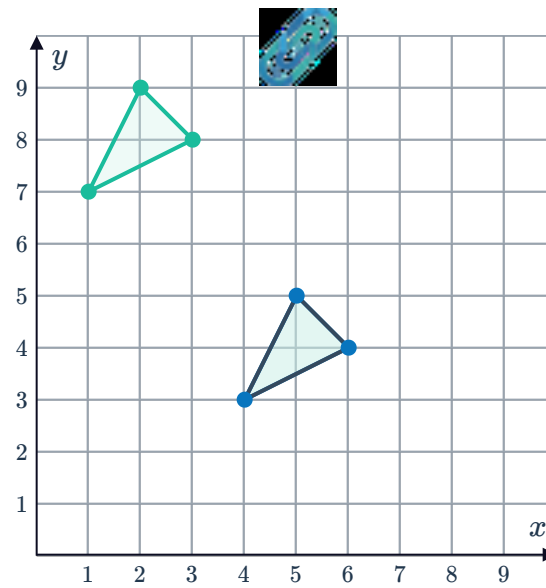
14

- a The preimage A has been translated 5 units down to produce the image A' .
- b The preimage $ABCD$ has been translated 6 units to the right to produce the image $A'B'C'D'$.
- c The preimage ABC has been translated left by 2 units and translated up by 1 unit to the image $A'B'C'$.
- d The preimage $ABCD$ has been translated right by 16 units and translated down by 12 units to the image $A'B'C'D'$.

15



16



17 (1,1)

18

a 8

b $c3, c5, d2, d6, f2, f6, g3, g5$

19 No. Block E and block N have the same vertical distance to the top of the tower, but different horizontal distances.